PROBABILITY

NECKER

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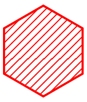
Contents

Symbols Legend

Throughout this document, the following graphical representations will be used to visualize sets and their relationships within formulas:

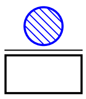
 **Universe (U):** Represents the total number of elements or possible cases (e.g., 100% of the class).

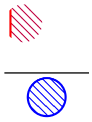
 **Event A:** The set of elements that satisfy the first condition (e.g., wearing glasses).

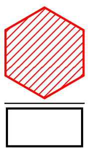
 **Event B:** The set of elements that satisfy the second condition (e.g., having red hair).

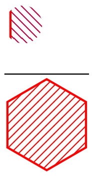
 **Intersection ($A \cap B$):** The set of elements that simultaneously satisfy both conditions (e.g., wearing glasses and having red hair).

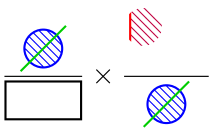
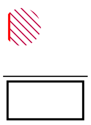
Universal Formulas

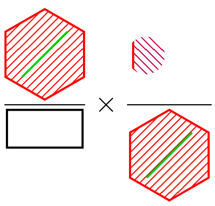
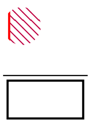
$P(A) =$ 

$P(B|A) =$ 

$P(B) =$ 

$P(A|B) =$ 

$P(A) \cdot P(B|A) =$  $=$ 

$P(B) \cdot P(A|B) =$  $=$ 

From which the classic formula derives:

$$P(A|B) = \frac{P(A) \cdot P(B|A)}{P(B)} \tag{1}$$

1 Combinatorics

Combinatorics deals with counting the number of possible groupings of objects within a finite set, given specific criteria. In other words, it allows determining how many different groups are possible given a construction rule.

1.1 Fundamental Counting Principle

If an experiment I has m outcomes and an experiment II has n outcomes, the total combinations are $m \times n$. Generalizing to k experiments:

$$\text{Total outcomes} = \prod_{i=1}^{k} m_i$$

1.2 Permutations (Order matters, AB and BA are NOT the same)

- **With repetition:** $D_{n,k} = n^k$. The same object can be chosen multiple times (n number of objects, k number of slots). E.g.: AA, AB, AC, BA
- **Without repetition:** $d_{n,k} = \frac{n!}{(n-k)!}$. Each object is unique. E.g.: AB, AC, BC, BA, CA, CB

1.3 Permutations of All Elements

A special case of permutations without repetition where all n objects are ordered: $P_n = n!$. (Thus, we do not have a limit of total slots, but we find all possible arrangements of all elements).

1.4 Combinations (Order does NOT matter, AB and BA are the same)

Only the groups themselves are considered. The number of combinations of n objects taken k at a time is given by the binomial coefficient:

$$C_{n,k} = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

2 Axiomatic Probability Theory

Probability quantifies uncertainty between a **subjective** meaning (personal belief) and a **frequentist** meaning (historical frequency).

2.1 Formalization with Set Theory

- **Universe Set (Ω):** Contains all possible outcomes (Certain Event).
- **Elementary Event (ω):** A single outcome (singleton).
- **Event ($E \subseteq \Omega$):** A subset of possible outcomes.

2.2 Algebra of Events

We must ensure that when combining them, the result is still an event we are capable of managing and measuring.
To say that a collection is "closed" under an operation means that the result of that operation never "falls" outside the algebra. Here is the deep meaning of the three properties that must hold for $\mathcal{A}$ to be an algebra:

1. **Closure under the Certain Event ($\Omega \in \mathcal{A}$):** The certain event represents everything that can happen (100% of the possibilities). An algebra must necessarily contain the universe Ω because we must always be able to say: "Something will surely happen". Without the certain event, we would have no reference point for the probability scale.

2. **Closure under Complementarity ($E \in \mathcal{A} \Rightarrow \bar{E} \in \mathcal{A}$):** The complement $\bar{E}$ (or E^c) represents the **negation** of an event (everything that is not E). *Logical meaning:* If the algebra allows us to evaluate the probability that it rains (E), it must mandatory allow us to evaluate the probability that it does **not** rain ($\bar{E}$). If you can think of an event, you must be able to think of its opposite.

3. **Closure under Union ($E, F \in \mathcal{A} \Rightarrow E \cup F \in \mathcal{A}$):** The union represents the logical **OR** operation. *Logical meaning:* If we are able to manage the event "the number 2 comes out" and the event "the number 4 comes out", we must also be able to manage the combined event "2 **or** 4 comes out". This property allows us to build complex events by assembling simpler building blocks.

2.2.1 Sigma-Algebra:

While standard algebra guarantees closure for a finite number of operations (I can combine 10, 100, or a million events), Sigma-Algebra additionally adds: If you have an infinite sequence of events $E_1, E_2, E_3 \ldots$ (which you can list one after another), the Sigma-Algebra guarantees that their infinite union also belongs to the system.

2.3 Kolmogorov Axioms

Kolmogorov's axioms formally define a probability function P which must satisfy:

A1. $0 \leq P(E) \leq 1$.

A2. $P(\Omega) = 1$.

A3. For disjoint events ($E \cap F = \emptyset$): $P(E \cup F) = P(E) + P(F)$. (The probability of the union of mutually exclusive elements is equal to the sum of their individual probabilities).

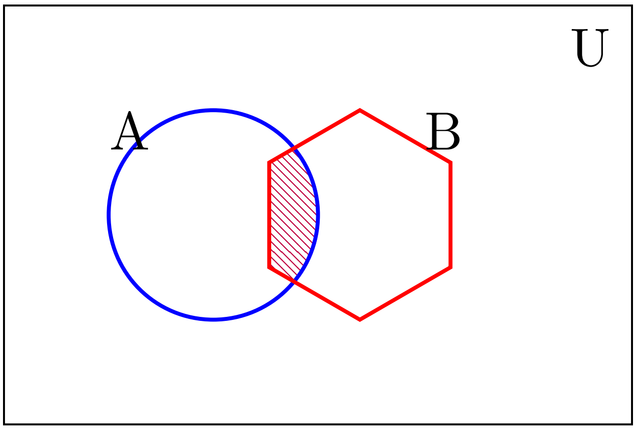
3 Conditional Probability

Conditional probability answers the question: *"What is the probability that event B occurs, knowing that event A has already occurred?"*.

In visual terms, this means that our **Universe** U (the entire class) disappears and is replaced by a **new, smaller Universe**: set A.

3.1 The Intersection

Before discussing conditional probability, we must define the intersection.
The intersection $A \cap B$ represents the event in which both A and B occur simultaneously.



3.2 The Mathematical Definition

The probability of B conditioned on A is defined as:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \tag{2}$$

Formula analysis:

- **The Numerator ($P(A \cap B)$):** Represents our "lens" (the intersection). These are the only cases of B that survive, because they are the only ones that also lie inside A.
- **The Denominator ($P(A)$):** It is our new world. By dividing by $P(A)$, we are declaring that anything outside of A no longer interests us.

`To understand better, by dividing by P(A) we are saying: how many times does the intersection fit inside A? Thus obtaining focus exclusively on A.`

3.3 The "Magnifying Glass" Effect

When we condition a probability, we are performing a **Zoom IN**. Imagine we have an intersection that occupies 5% of the entire class ($P(A \cap B) = 0.05$).

1. If we look at the entire class (100 people), the lens is small.
2. If we Zoom IN on A (which is only 20 people), that exact same lens now occupies 25% of the visible space.

$$P(B|A) = \frac{5\% \text{ (slice)}}{20\% \text{ (new cake)}} = 0.25 \rightarrow 25\%$$

3.4 The Product Rule

From the conditional probability formula derives the formula to calculate the intersection, which we used for the **Zoom OUT**:

$$P(A \cap B) = P(A) \cdot P(B|A) \tag{3}$$

This formula tells us that to find the "weight" of the intersection on the total system, we must take the conditional probability and "scale" it by the size of the group it belongs to.

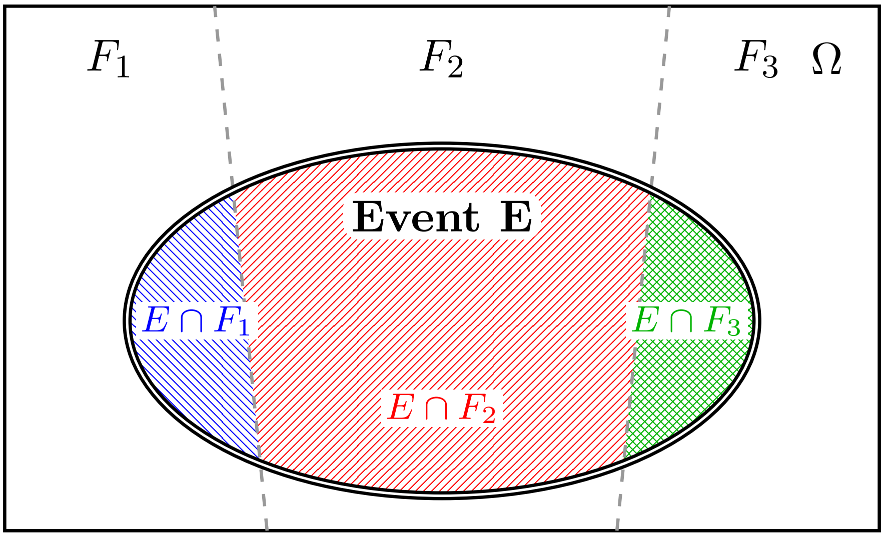
`To understand better, by multiplying by P(A) we are saying: we know P(B) in a universe of only A; to adapt it to the total universe, we must multiply it by how likely it is that picking a random element, that element belongs to A.`

4 Law of Total Probability

It allows calculating $P(E)$ by fragmenting it over a partition of Ω $(F_1, \ldots, F_n)$:

$$P(E) = \sum_{i=1}^{n} P(E|F_i)P(F_i)$$

Logically, we sum all the probabilities of the event occurring relative to its section to reconstruct the event E in its entirety over the total system.



5 Bayes' Theorem

Bayes' Theorem allows updating the probability of an event (called the prior) **like the initial possibility that a suspect is guilty among one hundred people**, transforming it into a conditional probability (called the posterior) when new evidence emerges, thereby executing a radical shift of focus that restricts the frame from the entire system to only the group of elements that exhibit that specific characteristic.

5.1 Practical Example: Glasses and Red Hair

Imagine a class of **100 people** (Universe U):

- **Event A:** Wearing glasses (20 people). $P(A) = \frac{20}{100} = 0.2$.
- **Event B:** Having red hair (10 people). $P(B) = \frac{10}{100} = 0.1$.
- **Intersection ($A \cap B$):** 5 people have both glasses and red hair.

5.2 Difference between Conditional and Bayes:

1. **Conditional ($P(B|A)$):** "If I look only *among those who wear glasses*, what probability do I have that the person has red hair knowing how many have both red hair and glasses?"

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{5}{20} = 25\%$$

(Our world has narrowed down to the 20 wearing glasses. The intersection carries significant weight: 1 out of 4).

2. **Bayes ($P(A|B)$):** "If I now look only *among those who have red hair*, what probability do I have that they wear glasses?"

$$P(A|B) = \frac{P(A) \cdot P(B|A)}{P(B)} = \frac{5/100}{10/100} = \frac{5}{10} = 50\%$$

(With Bayes, the necessity of knowing the event intersection probability explicitly is eliminated through the inverse formula of conditional probability).

5.3 Formula Construction: The Theorem as Logical Stratification

Let's return to the class example of 100 people to break down Bayes' Theorem and see how each operation adds fundamental piece of information until the final structure is completed.

1. **The Starting Point (The "zoomed" data):** We begin with $P(B|A) = 25\%$. *What do we know?* We only know the proportion of red-haired people inside those who wear glasses. We do not yet know how many people are in the total system.

2. **Integrating the Weight (Phase 1 - Zoom OUT):** We add to the calculation the probability of the origin group out of the total, $P(A) = 20\%$.

$$P(B|A) \cdot P(A) = 0.25 \cdot 0.20 = 5\%$$

What did we add? We added **scale to the system**. Now it represents the actual intersection ($P(A \cap B)$), namely the 5 physical people out of the system of 100.

3. **The Scenario Shift (Phase 2 - Zoom IN):** Now we introduce the new reference "universe", $P(B) = 10\%$. To do this, we divide the previous result by this value.

$$\frac{P(B|A) \cdot P(A)}{P(B)} = \frac{5\%}{10\%} = 50\%$$

What did we add? We added the **new focus**. By dividing, we "discard" everything that is not B and force our framing to shrink solely onto the group of red-haired people.

Final Result: Assembling these pieces, we arrive at the complete formula:

$$P(A|B) = \frac{\overbrace{P(B|A) \cdot P(A)}^{\text{Intersection Construction (Zoom OUT)}}}{\underbrace{P(B)}_{\text{Imposition of the new Focus (Zoom IN)}}} \tag{4}$$

6 Naive Bayes Classifier

The concept of conditional probability is the base of **classification**: given an object with certain characteristics (features), what is the probability that it belongs to class y_k?

6.1 The Real Problem: The Complexity of Intersections

Without any simplification, if we had multiple characteristics (e.g., Eye Color X_1 and Hair Color X_2), to calculate the probability that an individual belongs to the Marvel class (M), we would have to use Bayes like this:

$$P(M|X_1 \cap X_2) = \frac{P(X_1 \cap X_2|M) \cdot P(M)}{P(X_1 \cap X_2)}$$

Here arises the **computational problem**:

- If X_1 has m variants and X_2 has n, the intersection $P(X_1 \cap X_2|M)$ requires knowing the frequency of every single possible combination ($m \times n$).
- With hundreds of features, the number of combinations explodes exponentially, making it impossible to have enough data in the dataset to estimate each specific intersection.

6.2 The "Naive" Assumption

To solve this, the Naive Bayes classifier introduces a drastic simplification: it assumes that, given the class, the features are **independent** of each other. *Logic:* We assume that hair color does not influence eye color once we know the character belongs to Marvel.

Thanks to this assumption, the complex intersection (the logical product) transforms into a simple product of individual probabilities:

$$P(X_1 \cap X_2 \cap \cdots \cap X_n|Y) \approx P(X_1|Y) \cdot P(X_2|Y) \cdot \cdots \cdot P(X_n|Y)$$

We move from an **exponential** calculation cost ($m \times n \times \dots$) to a **linear** one ($m + n + \dots$).

6.3 Why the Denominator Disappears

In Bayes' formula, the denominator $P(X_1 \cap \cdots \cap X_n)$ represents the probability of observing that specific combination of features across the entire dataset.

- When we must decide whether an individual is "Marvel" or "DC", we compute the probability for both classes.
- We notice that the denominator is **identical** for all classes, since it depends solely on the features of the individual in front of us, not on the class we are testing.
- Being a constant that does not alter the relative strength between results, we can ignore it for classification purposes.

6.4 The Decision Rule: The Argmax

Since we have removed the denominator and simplified the numerator, the classifier no longer searches for the exact probability value, but instead searches for **the most probable class**. The **arg max** operator (argument of the maximum) is therefore utilized:

$$\hat{y} = \arg\max_k \left[P(y_k) \prod_{i=1}^{n} P(X_i|y_k) \right] \tag{5}$$

What does it mean?

- $P(y_k)$: The *prior* probability of the class (e.g., how many Marvel characters there are in total).
- $\prod P(X_i|y_k)$: How well the features of the individual "fit" class k.
- **arg max**: Means "return the index k (the class) that generated the highest value". We do not need to know if the probability is 0.7 or 0.8; we just need to know which class wins.

7 Independent Events

Two events are independent ($E \perp F$) if knowing one does not influence the other:

$$P(E|F) = P(E) \iff P(E \cap F) = P(E) \cdot P(F)$$

Properties: If $E \perp F$, then $E \perp \bar{F}$. The independence of a set of events requires that the factorization holds for every possible subset of the intersection.